A Unified Attention Paradigm for Hyperspectral Image Classification

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***Abstract*— Attention mechanisms improve the classification accuracies by enhancing the salient information for hyperspectral images (HSIs). However, existing HSI attention models are driven by advanced achievements of computer vision, which are not able to fully exploit the spectral–spatial structure prior of HSIs and effectively refine features from a global perspective. In this article, we propose a unified attention paradigm (UAP) that defines the attention mechanism as a general three-stage process including optimizing feature representations, strengthening information interaction, and emphasizing meaningful information. Meanwhile, we designed a novel efficient spectral–spatial attention module (ESSAM) under this paradigm, which adaptively adjusts feature responses along the spectral and spatial dimensions at an extremely low parameter cost. Specifically, we construct a parameter-free spectral attention block that employs multiscale structured encodings and similarity calculations to perform global cross-channel interactions, and a memory-enhanced spatial attention block that captures key semantics of images stored in a learnable memory unit and models global spatial relationship by constructing semantic-to-pixel dependencies. ESSAM takes full account of the spatial distribution and low-dimensional characteristics of HSIs, with better interpretability and lower complexity. We develop a dense convolutional network based on efficient spectral–spatial attention network (ESSAN) and experiment on three real hyperspectral datasets. The experimental results demonstrate that the proposed ESSAM brings higher accuracy improvement compared to advanced attention models.**

***Index Terms*— Attention mechanism, deep learning (DL), hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL remote sensing integrated spectroscopy and imaging technology realizes the simultaneous acquisition of spectral information and spatial images, which has become a significant breakthrough in the field of remote sensing. Unlike RGB images with only several bands, hyperspectral images (HSIs) allow each pixel to be abstracted as a continuous curve containing hundreds of bands, which presents spectral properties of the land cover and reflects fine distinctions between different categories. HSIs address the need to detect the composition and morphological characteristics of target objects, promoting the wide range of applications such as precision agriculture [1], geological survey [2], environmental monitoring [3], and biotechnology [4]. HSI classification, a vibrant research subject of hyperspectral remote sensing, aims to allocate a unique class label to each pixel based on its spectral information and faces challenges from the high dimensionality, redundant information, and limited labeled samples [5]. Classification methods designed for natural images are not necessarily suitable for HSIs although advances in HSI classification are driven by the computer vision. Therefore, a series of classification methods has been developed specifically for hyperspectral data in recent years.

The essence of HSI classification is to seek a mapping function from the input image data to the output category labels, with the goal of minimizing the residual between the desired output and the actual output [5]. Further, it can be decomposed into two parts with the feature transformation and the category discriminant. In this regard, classifying HSI can be mathematically formulated as follows:

$$Y = f_{\mathrm{d}}(f_{\mathrm{t}}(X, \theta_{\mathrm{t}}), \theta_{\mathrm{d}}) \tag{1}$$

where $X \in \mathbb{R}^{H\times W\times B}$ and $Y \in \mathbb{R}^{H\times W\times L}$ denote the origin data of HSI scene and the one-hot encoding of corresponding labels, respectively. H, W, B, and L are its height, width, the number of bands, and the number of classes, respectively. The transformation function f_{t} (with or without parameters θ_{t}) converts spectral information to the discriminative representation, which is projected as the label encoding by the discriminant function f_{d} (with or without parameters θ_{d}). A central theme of HSI classification is capturing more powerful representations to reflect most salient properties of pixels.

Over the past decade, researchers have proposed a range of HSI classification methods, from the traditional approaches to the current deep neural network models. Traditional methods abstract spectral or spatial information representing various image attributes from hyperspectral data through feature transformation methods such as band selection, principal component analysis (PCA), morphological operations [7], and local binary patterns (LBPs) [8]. These discriminative features are delivered to classical classifiers like support vector machine (SVM), K-nearest neighbors (KNNs), and sparse representation (SR) [9] to seek the best boundaries of the feature space. Considering that HSIs are intrinsically 3-D tensors,

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some traditional methods treat hyperspectral data as a tensor cube and develop a range of tensor-based methods to exploit the inherent structure of HSIs [10], [11]. Although traditional methods can effectively represent and analyze hyperspectral data, manual design features rely on expert knowledge and cannot adapt to the differences between different real data [12]. Different from traditional methods, deep learning (DL) techniques provide a data-driven learning strategy that allows adaptively extract discriminative high-level features from raw data by stacking successive hierarchical systems. DL methods for HSI classification exploit neural networks as feature extraction tools to obtain discriminative high-level sematic features and achieve excellent performances. Cheng et al. [13] developed a unified framework that extracts more powerful spectral–spatial features via off-the-shelf convolutional neural network (CNN) models and SVM with the metric learning regularization. Zhou et al. [14] proposed a stacked autoencoder (SAE) framework with the local Fisher discriminant regularization to acquire compact and discriminative low-dimensional feature mappings. In addition, Hong et al. [15] developed graph convolutional networks (GCNs) combined with CNNs and provided more discriminative multiperspective feature representations.

Among aforementioned neural networks, CNN integrates spectral features with spatial neighborhood information in an effective and flexible manner and becomes a main part of HSI representation tools. Common CNNs can be divided into patch-based and image-based approaches depending on their input patterns. Patch-based methods treat processing pixels as independent individuals, which extract pixels' neighborhood cubes of a fixed size as inputs and perform inference and categorization in a pixel-by-pixel manner. One family of such approaches adopts the idea of divide-and-conquer to process spectral and spatial features in parallel, typical of which is the dual-branch CNNs to exploit 1-D CNNs for spectral vectors of pixels and 2-D CNNs for the spatial contextual information [16], [17]. Another design idea is to adopt 3-D CNNs for simultaneous extraction of spectral and spatial information [18], [19]. In patch-based methods, the target pixel is endowed a specific label by all pixels of its neighborhood, which leads to lack the holistic understanding of HSIs and suffers from computational redundancy due to overlapping neighborhood patches. Image-based approaches based on the fully convolutional network (FCN) framework regard the full HSI as input. Xu et al. [20] combined a two-branch structure with FCN to establish a spectral–spatial FCN. Cui et al. [21] built a multiscale spectral–spatial FCN with feature pyramid to expand the receptive field and strengthen feature diversity. In addition, Zheng et al. [22] and Zhu et al. [23] adopted encoder–decoder FCNs to share the computation in the spatial dimension and explore the global spatial information. The FCNs implement the fast full-image inference with stacked convolutional layers and up-and-down sampling strategy, which expand the scope of spatial relationship modeling to improve the quality of the representation but ignore deeper exploration of the intrinsic structure of HSIs.

The representation capability of CNNs is influenced by the amount of training data and the network architecture. The former is difficult to obtain for HSIs; hence, one family of CNNs is trained in a semisupervised manner by adding unlabeled samples with a high degree of positivity to enhance the generalization ability [24]. The latter needs to tradeoff between the size of parameters and classification performance. Therefore, adding attention modules to existing models becomes popular, as it brings significant performance gains at slight additional model parameters. Attention mechanism was inspired by the phenomenon that human prioritizes salient regions in complex scenes and introduced into computer vision with the aim of further emphasizing informative features and ensuring the effectiveness of deep features [25]. Frontier works, which leveraged more efficient attention modules to assist CNNs, have demonstrated the benefit of attention mechanisms for providing the higher quality of representations and superior classification results. Mou and Zhu [26] designed the spectral attention module to aggregate spectral–spatial contexts and recalibrate strengths of different bands. Inspired by the classic squeeze-and-excitation network (SENet), references [27], [28], and [29] introduced the spectralwise attention modules with the global pooling operation and the double-layer fully connected structure to capture interband correlations and reallocate weight responses in spectral dimension. Alongside spectral attention modules that emphasize useful bands and suppress redundant ones, spatial attention mechanisms were also applied to HSI classification to enhance discrimination for complex land covers by distinguishing adjacent regions that are significantly correlated with target pixels. The most representative way is self-attention manner [30] to obtain the relationship map between the target pixel and neighboring pixels and reconstruct the pixel features via weighted summation [31], [32], [33]. To seek for simultaneous feature refinement in spectral and spatial dimensions, references [34], [35], [36], and [37] gather significant information in two separate dimensions in the form of self-attention. Analogously, references [23] and [38] introduce the convolutional block attention module (CBAM) [39] that performs the information interaction along spectral and spatial domains through fully connected layers and convolutional layers to guide feature reoptimization.

Attention mechanisms achieve information interaction and selection to predict "which" bands and "where" features are of interest, and bring clear performance gains for HSI classification. However, we notice still some potential issues to be explored in this community.

1) Soft-attention mechanisms, such as SENet and CBAM, adopt the single statistic of feature maps as the global representation to construct long-range dependencies. This type is certainly concise but may cause inadequate expression of feature maps.
2) Attention mechanisms use cross-channel or cross-pixel interactions to construct long-range relationships of HSI. Attention modules in patch-based models cannot compensate for the lack of global dependence modeling in CNNs. However, directly establishing the one-to-one

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correspondence in the whole HSI faces the limitation of high computational cost.

3) HSIs exhibit the unique structural properties of high-dimensional data. Generic models for natural images may not fully consider the spectral–spatial prior and structure of HSIs.

The above issues give us an incentive to rethink the attention mechanism for HSI and explore a unified attention paradigm (UAP) with the corresponding model framework, which divided the process of the attention mechanism into three distinct parts. In this foundation, we design an efficient spectral–spatial attention module (ESSAM) adapted to the unique structure of hyperspectral data, achieving a balance of performance and complexity while fully considering the spectral characteristics and spatial prior, with batter interpretability and generalization capability. The main contributions of this article are summarized as follows.

1) We propose a unified paradigm with three stages that fully reveals the working principle of the attention mechanism. Specifically, UAP abstracts a globalized representation of each feature map, on the basis of which a global relationship map is generated to guide cross-information interaction, and then adaptively relocates or reassigns crucial features. This paradigm not only explains that the key to the attention mechanism lies in information induction, relational reasoning, and feature redistribution, but also provides clear entry points for the improvement of attention models.
2) We concretize the UAP and design a parameter-free spectral attention (PFSA) block. PFSA block exploits multiscale spatial averages as global spatial representations, alleviating the global spatial feature representation difficulty for different scales of land covers. At the same time, similarity calculation is used instead of parameter learning for global relationship inference to achieve parameter-free prediction of significant spectral bands. PFSA block fully takes into account the spatial prior of regional distribution, and its internal process is clear and explainable.
3) Based on the assumption that pixels in the same class of HSIs lie in the same low-dimensional subspace, the memory-enhanced spatial attention (MESA) block is proposed to approximate latent representations of HSIs using the learnable memory unit (LMU). In MESA blocks, long-range dependencies between all pixels are implicitly constructed to achieve efficient screening of key spatial locations at slight parameter cost. MESA block takes into account the sparsity of high-dimensional data and provides a balance of computational cost and classification performance.

To provide evidence for these claims, we develop a dense convolutional network based on efficient spectral–spatial attention network (ESSAN) and experiment on three public HSI datasets to demonstrate the superiority of our model over other state-of-the-art (SOTA) classification methods.

The remainder of this article is organized as follows. Section II briefly reviews the classic attention modules. Section III describes the proposed UAP and the frameworks of PFSA and MESA blocks. The details and the discussion of the experimental results are provided in Section IV. Finally, Section V concludes this article.

II. Classic Attention Modules

A. Soft-Attention Mechanism

Each channel of features corresponds to a convolutional kernel for CNNs, and different channels denote different semantic information [40]. The process of recalibrating channel weights by channel attention can be considered as a rescreening of semantic features to decide "what" the network should focus on more. By analogy to HSI classification, channel attention determines "which" spectral bands to attend, called spectral attention. Hu et al. [27] first proposed the concept of channel attention and instantiated it as the SE block, as shown in Fig. 1(a). The SE module is divided into two parts: squeeze and excitation, where the former generates the global spatial information of each channel via global averaging pooling operator and the later captures channelwise relationships by the combined operation of fully connected and nonlinear layers. Attention vector obtained through above steps are multiplied with the input features to complete the refinement of the original information in the channel dimension. Assuming that $X \in \mathbb{R}^{H\times W\times C}$ is the input features and $Y \in \mathbb{R}^{H\times W\times C}$ is the output (C is the number of channels), the computational process of the SE block can be expressed as

$$F_{\mathrm{s}} = \mathrm{GAP}(X) \tag{2}$$

$$F_{\mathrm{e}} = \mathrm{MLP}(F_{\mathrm{s}}) \tag{3}$$

$$Y = \mathrm{Sigmoid}(F_{\mathrm{e}}) \odot X. \tag{4}$$

Here, GAP denotes global average pooling in corresponding domain. MLP consisting of a two-layer perceptron provides the fully connected mapping. Sigmoid is the sigmoid function. $\odot$ is the Hadamard product.

The SE block is concise and plug-and-play in design. However, the global average pooling operation is too simple to adequately capture complex spatial structure, while the successive fully connected calculations for channel dependence learning increase the complexity of the model. A number of subsequent studies [41], [42], [43], [44], [45] have attempted to improve SENet from different perspectives, among which deep convolutional neural network with the efficient channel attention model (ECANet) [41] reduces the complexity of the model by replacing the continuous fully connected layers with a 1-D convolution layer for KNN cross-channel relationship learning, as shown in Fig. 1(b). The computational procedure of the efficient channel attention (ECA) module only modifies (3) of the SE block as follows:

$$F_{\mathrm{eca}} = \mathrm{Conv1D}(F_{\mathrm{s}}) \tag{5}$$

where Conv1D means the 1-D convolutional operation.

Channel attention determines "what" to pay attention to, while spatial attention focuses on "where" to attend. CBAM [39] solves both of these problems simultaneously with sequentially connected channel and spatial attention modules.

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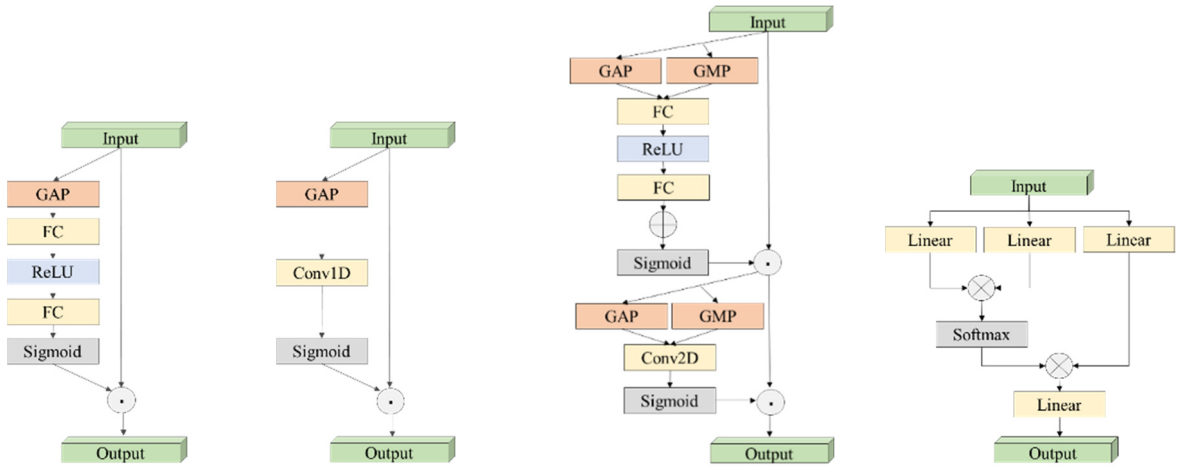

(a) (b) (c) (d)

Fig. 1. Classic attention models: (a) SE block; (b) ECA block; (c) CBAM; and (d) nonlocal module.

The structure diagram of CBAM is shown in Fig. 1(c). Formally, it can be expressed as

$$F_{\text{ca}} = \text{MLP}(\text{GAP}(X)) + \text{MLP}(\text{GMP}(X)) \tag{6}$$

$$X_{\text{ca}} = \text{Sigmoid}(F_{\text{ca}}) \odot X \tag{7}$$

$$F_{\text{sa}} = \text{Conv2D}([\text{GAP}(X_{\text{ca}}), \text{GMP}(X_{\text{ca}})]) \tag{8}$$

$$Y = \text{Sigmoid}(F_{\text{sa}}) \odot X_{\text{sa}} \tag{9}$$

where GMP is the global max pooling operation and Conv2D means the 2-D convolutional operation. Compared to SE block, the channel attention of CBAM adds the global max pooling feature in corresponding domain assisting infer finer channelwise attention. Spatial attention similarly encodes on average-pooled and max-pooled features to obtain the attention map, where the relational inference is realized by a large-size 2-D convolution operation and can be regarded as KNN correlation learning in the spatial domain. CBAM achieves significant information search in both channel and spatial dimensions in a lightweight manner, where the average-pooled and max-pooled information aggregation enhances the optimization capability of the module. However, the range of relationship modeling in the spatial domain is limited by the size of the convolution kernel.

B. Self-Attention Mechanism

The nonlocal network [46] introduces the self-attention into computer vision to capture contextual information at arbitrary distances with pairwise computation of spatial locations, which enhances the representation capability with a global perspective and complements CNNs for local spatial modeling. Its architecture is shown in Fig. 1(d), and the corresponding computational process is represented as

$$Q, K, V = \text{Linear}(X) \tag{10}$$

$$M = \text{Softmax}(QK^{\text{T}}) \tag{11}$$

$$Y = MV. \tag{12}$$

Here, $X \in \mathbb{R}^{H\times W\times C}$ and $Y \in \mathbb{R}^{H\times W\times C'}$ are the input and output features, respectively. Self-attention mechanisms, represented by the nonlocal module, first generate the queries, keys, and values $Q, K, V \in \mathbb{R}^{HW\times C'}$ by linear transformations and reshaping operations. The queries indicate which feature vectors to focus on, which are used to calculate with the keys to obtain importance scores of vectors in K. Softmax is the softmax function to redistribute importance scores. $M \in \mathbb{R}^{HW\times HW}$ is the relationship map consisting of importance scores, which provides an intuitive indication of the importance of each feature vector relative to the other feature vectors [47]. The output of self-attention is produced by a weighted average of values V with the relationship map. In contrast to the soft attention modules described above, the relationship map of the self-attention model provides a rather intuitive interpretation, which allows for a deeper understanding of the model behavior, and the relations between inputs and outputs [47]. However, the complexity of the self-attention is $O((H \times W)^2)$. Computational demands in both memory and time make it impractical to extend self-attention mechanisms on large-size images.

Many works have improved the nonlocal network from the perspective of reducing the computational complexity. Criss-cross network (CCNet) [48] splits spatial attention into row attention and column attention and implements global interaction by recurrent relational reasoning. Expectation-maximization attention network (EMANet) [49] abandons computations of densely connected relation graph and instead iterates a set of compact bases by the expectation-maximization (EM) algorithm, which implicitly constructs global relations with descriptors of the whole image and greatly reduces the complexity. In addition, there are also object-contextual representation network (OCRNet) [50], double attention network (A2Net) [51], global context network (GCNet) [52], etc., which optimize self-attention from multiple perspectives.

Currently, various types of attention mechanisms have been widely used in different computer vision tasks, showing notable results. Inspired by above studies, we try to propose a UAP and design a concise and efficient spectral–spatial attention model adapting to high spectral resolution, complex spatial distribution, and limited labeled samples of HSIs.

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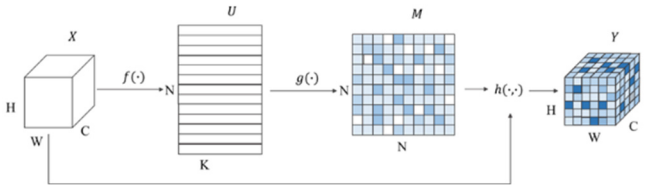

Fig. 2. Unified attention framework.

III. Proposed Framework

A. Unified Attention Paradigm

Given the input feature is $X \in \mathbb{R}^{H\times W\times C}$, where H, W, and C correspond to the height, width, and the number of channels, respectively, the output feature $Y \in \mathbb{R}^{H\times W\times C}$ can be generated by the three-stage UAP consisted of information abstraction, cross-location interaction, and salient information fusion. Viewing each stage as a mapping function, the overall form can be formulated as

$$Y = h(g(f(X)), X). \tag{13}$$

The corresponding framework is illustrated in Fig. 2. Specifically, f is responsible for abstracting a concise and discriminative global representation U of the input features by corresponding dimensions, which serve for subsequent relational inference. g implements cross-information interactions supported and generates the relational matrix M to capture long-range dependencies. h aggregates relevant information guided by the graph M to adaptively distinguish the significant regions and enhance the feature representation.

With the above paradigm, it is not difficult to find that either soft-attention, self-attention, spectral attention, or spatial attention can be split into the three stages mentioned. Equations (2)–(4) of SE block and (10)–(12) of the nonlocal module can correspond to f, h, and g in (13). Soft-attention modules mostly adopt global pooling or convolution operation as f. Its relation modeling is implemented by fully connected layers or the convolutional layer, where the relation matrix M is continuously optimized by backpropagation in the form of network parameters. h reconstructs the input descriptor of each channel or pixel, which be mapped to a set of the attention weights and directly multiply with the input to implement dynamic selection. Self-attention utilizes convolutional layers as f to achieve linear transformation of the original input and takes feature multiplications as g to capture internal similarity. It explicitly calculates the relation matrix M that is used to weighted reconstruct transformed features to achieve feature enhancement. Therefore, the specific attention module can be adapted to design different network structures for functions f, g, and h and meet the needs of realistic tasks.

It is not hard to find that the key of the whole attention model lies in the construction of functions f and g. The function f is the basis, where the abstraction of input should be robust and efficient. The overly simple representation will reduce the enhancement effect, while the high-dimensional representation will increase the computational cost of the whole model. The relational inference range of function g, whether in the form of network parameters or explicitly computed to capture internal dependencies, directly affects the complexity and effectiveness of the model. We instantiate f and g according to the task difficulty and data structure prior of HSI classification, and propose the PFSA block and MESA block, respectively.

B. Parameter-Free Spectral Attention

Most spectral attention methods construct interband correlations via global spatial descriptor of each spectral channel, thus causing complex spatial structures of HSIs cannot be adequately represented. Considering that land covers of the same category are likely to appear in clusters in HSIs, we introduce multiscale spatial descriptors into the spectral attention to integrate spatial statistics of different granularities and provide the concise and high-quality spectralwise representation. Specifically, we uniformly divide the input feature $X \in \mathbb{R}^{H\times W\times C}$ into multiple feature patches according to different spatial scales k_1 and k_2. In other words, X is repeatedly cut twice in the spatial dimension to obtain k_1^2 and k_1^2 nonoverlapping subfeature maps, respectively. We aggregate all feature patches to form a set $S = \{S_i\}_{i=1}^{K}$, where $K = k_1^2 + k_2^2$ is the total number of subfeature maps. Assuming that $x_j^i \in \mathbb{R}^C$ denotes the spectral information of the jth pixel in S_i, we can obtain the average of each subfeature map as the center to represent spatial contextual information

$$s_i = \frac{1}{N_i}\sum_{j=1}^{N_i} x_j^i. \tag{14}$$

Here, N_i denotes the total number of pixels in S_i, and s_i means the center of S_i. The multiscale descriptors $U = [s_1, s_2, \ldots, s_K]$ of all bands are generated by aggregating all centers, where $U \in \mathbb{R}^{C\times K}$. For each band, there is a K-dimensional vector to represent its multiscale spatial contextual information. Therefore, we can define the function f from the input to the channelwise representation as

$$f(X) = \left[\frac{1}{N_1}\sum_{j=1}^{N_1} x_j^1, \frac{1}{N_2}\sum_{j=1}^{N_2} x_j^2, \ldots, \frac{1}{N_K}\sum_{j=1}^{N_K} x_j^K\right]^{\mathrm{T}}. \tag{15}$$

We employ a collection of spatial average pooling operations at different scales to achieve (15). k_1 and k_2 are hyperparameters of pooling layers.

In response to the high spectral dimensionality of hyperspectral data and the lack of trainable data, we hope instantiate spectral attention is low complexity and can learn a wider range of cross-band relationships with as few parameters as possible. Dot product operation is an idea candidate, as it is parameter-free global interaction, while its validity is verified in self-attention. In this stage, we can obtain a densely connected graph $M \in \mathbb{R}^{C\times C}$ preserving correlation between arbitrary bands

$$M_{i,j} = \frac{e^{U_i U_j^{\mathrm{T}}}}{\sum_{c=1}^{C} e^{U_i U_c^{\mathrm{T}}}} \tag{16}$$

where U_i is the multiscale representation of ith spectral channel, and i and j are the channel subscripts. Therefore,

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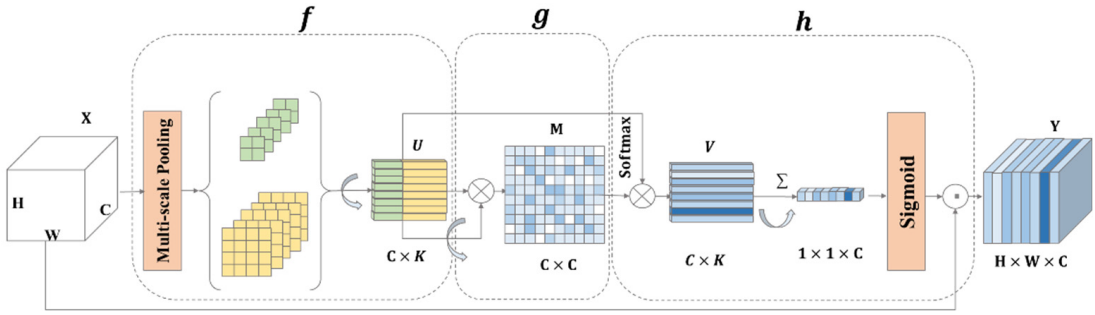

Fig. 3. PFSA block.

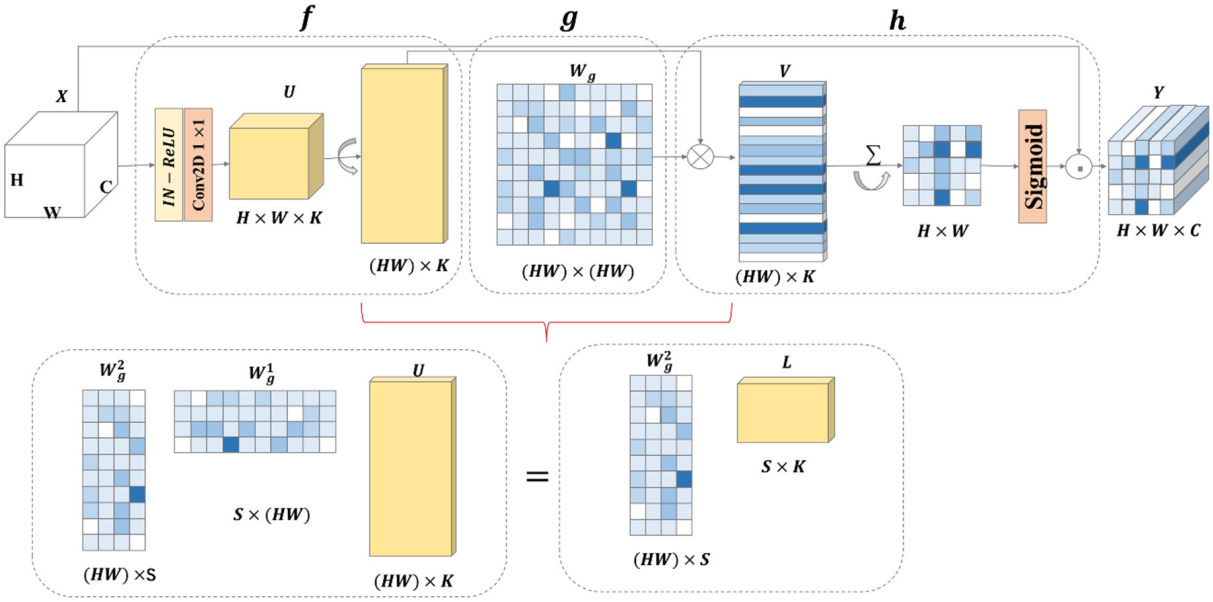

Fig. 4. Assumed spatial attention block.

the function g of PFSA can be written as

$$g(U) = \text{Softmax}\left(UU^{\text{T}}\right). \tag{17}$$

The final output can be generated by information fusion and readjustment of function h

$$h(U, M, X) = \text{Sigmoid}\left(\sum_{K} MU\right) \odot X \tag{18}$$

which is directed by M to calculate and activate the spectral attention vector.

The network structure of PFSA can be constructed according to (15), (17), and (18), as shown in Fig. 3. Compared with most channel attention mechanisms, PFSA block adaptively aggregates multigranularity semantic information with multiscale pooling to improve the robustness of spectral representation, while the parameter-free relational modeling can fully capture band dependencies and dynamically adjust the spectral channel weights in a simple and flexible manner.

C. Memory-Enhanced Spatial Attention

We try to extend PFSA to the spatial dimension and generate a spatial attention map for the selection of focal regions, as shown in Fig. 4. Unlike PFSA block, the abstracted pixel representation can be achieved by nonlinear mapping to dimensionality reduction and feature transformation synchronously. Thus, the mapping function f transforming the input features $X \in \mathbb{R}^{(HW)\times C}$ into the spatial representation $U \in \mathbb{R}^{(HW)\times K}$ can be written in the following form:

$$f(X) = \delta(W_{\text{f}}X) \tag{19}$$

where $W_{\text{f}} \in \mathbb{R}^{C\times K}$ and δ refers to the combination of instance normalization (IN) and the rectified linear unit (ReLU). This stage can be implemented by a 1×1 2-D convolutional layer. Since even small-size images contain thousands of pixels, the computational cost associated with direct pairwise calculations in the function g is unaffordable. The most straightforward replacement solution is to perform global spatial relationship inference with learnable parameters, i.e., the relation matrix M is stored in W_{g}. And the function h can be formulated as

$$h\left(U, W_{\text{g}}, X\right) = \text{Sigmoid}\left(\sum_{K} W_{\text{g}}U\right) \odot X. \tag{20}$$

Here, $W_{\text{g}} \in \mathbb{R}^{(HW)\times(HW)}$, and it brings a large number of parameters, which will increase the model complexity dramatically and is fatal to the HSI classification task.

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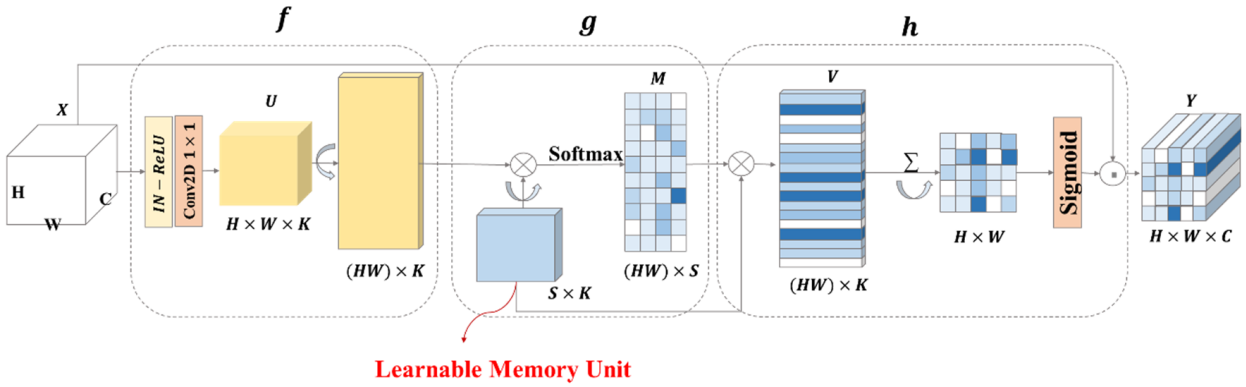

Fig. 5. MESA block.

Inspired by the bottleneck structure of SE block, we split W_g into two low-dimensional matrices $W_g^1 \in \mathbb{R}^{(HW)\times S}$ and $W_g^2 \in \mathbb{R}^{(HW)\times S}$ to compress the number of parameters and limit the model complexity. Thus, the function h can be rewritten as

$$h\left(U, W_g^1, W_g^2, X\right) = \text{Sigmoid}\left(\sum_K W_g^2(W_g^1 U)\right) \odot X. \quad (21)$$

For HSIs, the spectral information of pixels belonging to the same class is highly correlated, and thus, these pixels are considered to lie in the same low-dimensional manifold. We can assume that each pixel exists a compact representation about a specific set of bases, which encodes the semantic information of the image. If this set of bases can be derived, the effective semantics of HSIs can be preserved to the maximum extent. Analyzing (21) again from this perspective, we can approximate that $W_g^1 U$ is looking for valid semantic information, and W_g^2 can be regarded as a mapping matrix to recover pixel features with above semantic information. It establishes the relations from latent global semantics to individual pixels. Considering the iterative learning property of network parameters, we further set the global information descriptors as a set of trainable parameters $L \in \mathbb{R}^{S\times K}$, which are stored in LMU, and learn the semantic encoding adaptively using gradient backpropagation. Similar to PFSA, the process of calculating the relationship between pixels and descriptors can be written as

$$M_{i,j} = \frac{e^{U_i L_j^{\mathrm{T}}}}{\sum_{s=1}^{S} e^{U_i L_s^{\mathrm{T}}}} \quad (22)$$

where L_j is the jth descriptor, $1 \le j \le S$, and S is the total number of memory atoms and takes an integer multiple of the number of land cover classes. The function g can be formulated as follows:

$$g(U) = \text{Softmax}\left(U L^{\mathrm{T}}\right). \quad (23)$$

The function h first propagates global information to each pixel through recovering pixel features from global semantic, indirectly achieving information enhancement on a full spatial range, and then generates the spatial attention map acting on the input

$$h(M, X) = \text{Sigmoid}\left(\sum_K ML\right) \odot X. \quad (24)$$

The network structure of the above MESA can be constructed according to (19), (23), and (24), as shown in Fig. 5. Compared with CBAM, our scope of relationship modeling is not confined to the size of the convolutional filters with the same slight number of the required parameters. Compared with self-attention, MESA takes full advantage of the sparse property of HSIs to greatly improve the computational efficiency with better interpretability.

D. Efficient Spectral–Spatial Attention Network

In order to refine the classification feature from a spectral–spatial perspective, we arrange the PFSA and MESA submodules to compose an ESSAM and integrate it into a dense convolutional backbone network based on the FCN to construct the ESSAN. Its framework is illustrated in Fig. 6.

In our framework, a 1×1 2-D convolutional layer is the first layer before the densely connected structure to reduce dimensionality and avoid overfitting. The subsequent dense convolutional module plays the role of the initial spectral–spatial feature extraction. Then, ESSAM is applied for global feature refinement to boost the quality of representations for HSIs. In the training phase, we apply cross-entropy loss to optimize the parameters. Notice that HSI classification is a pixel-level classification task based on a single image, and the training and test samples exist in the same image. Hence, we only calculate the residual loss of the training samples in the training phase.

IV. Experimental Results

To evaluate our ESSAN on HSI classification, we carry out comparative experiments with other SOTA HSI classification CNNs on three of the most popular publicly available datasets, namely, Indian Pines, University of Pavia, and Kennedy Space Center. We employ the Adam optimizer with a learning rate of 0.0003 for 1000 training epochs, and empirically set the spatial scales of PFSA to [2] and [4] and the memory-atom number of MESA to twice the number of categories in each

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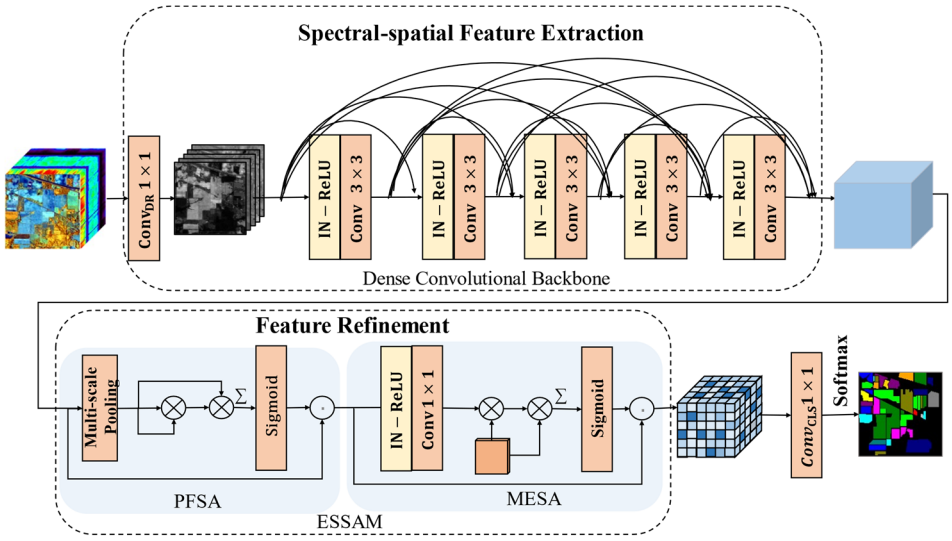

Fig. 6. Architecture of ESSAN.

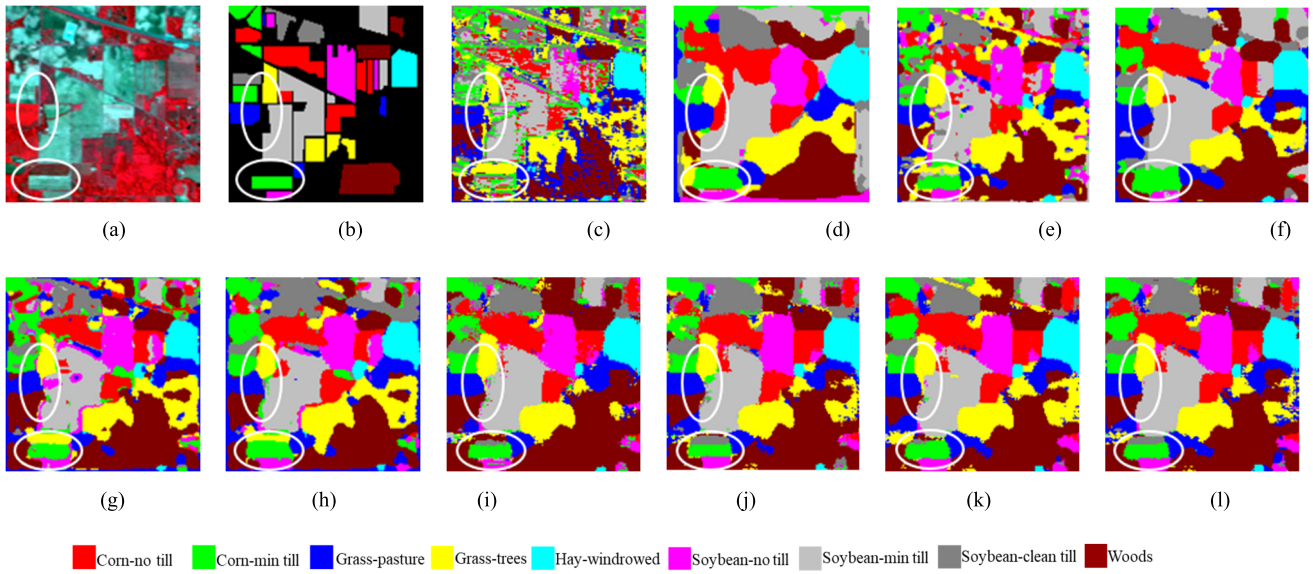

Fig. 7. Classification maps of different methods for Indian Pines dataset: (a) false-color image; (b) ground-truth map; (c) 3-DCNN; (d) DFFN; (e) SSRN; (f) FDSSC; (g) DBMA; (h) DBDA; (i) Baseline; (j) PFSAN; (k) MESAN; and (l) ESSAN.

dataset. Each group of experiments is repeated ten times and provides the means of three quantitative metrics, i.e., overall accuracy (OA), average accuracy (AA), and Kappa coefficient for performance evaluation. In addition, we also offer visual results of three datasets, which contain the classification of unlabeled background pixels as a reference for the generalizability of the models.

The comparison methods include: 3-DCNN [18], deep feature fusion network (DFFN) [53], spectral-spatial residual network (SSRN) [54], and fast dense spectral–spatial convolution network framework (FDSSC) [55] representing the most compared benchmarks, and double-branch multi-attention mechanism network (DBMA) [38] and double-branch dual-attention mechanism network (DBDA) [34] representing the SOTA attention-based methods. To facilitate observation of the effect of ESSAM, we construct the backbone network (Baseline) by removing ESSAM from ESSAN to generate Baseline results. In addition, we build the backbone network with PFSA (PFSAN) and the backbone network with MESA (MESAN) by retaining only the spectral or spatial attention components in ESSAM to verify their effect.

A. Experimental Comparison With Competitive Methods

1) Indian Pines: It was acquired by AVIRIS sensor over the Indian Pines test area in Northwest Indiana. The dataset contains 145 × 145 pixels and 224 reflectance bands with a spatial resolution of 20 m and a wavelength range of 400–2500 nm, of which 200 bands were retained after removing noise and absorption bands. The scene includes 16 types of land covers, of which we sampled and experimented with only nine classes with sufficient number of labeled samples. We randomly select 30 and 30 samples per class from 9345 labeled samples for training and validation, respectively. The specific category distribution is shown in Table I.

The quantitative and visual results are shown in Table II and Fig. 7, respectively. Compared with prior works, our ESSAN provides the highest overall accuracies on three overall metrics. At the same time, ESSAM shows the best

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TABLE I
LABELED SAMPLE STATISTICS FOR DIFFERENT DATASETS

No.	Indian Pines		University of Pavia		Kennedy Space Center	
	Class Name	Samples	Class Name	Samples	Class Name	Samples
1	Corn-no till	1434	Asphalt	6631	Scrub	716
2	Corn-min till	834	Meadow	18649	Willow swamp	243
3	Grass-pasture	497	Gravel	2099	CP hammock	256
4	Grass-trees	747	Trees	3064	Oak hammock	252
5	Hay-windrowed	489	Metal sheets	1345	Slash pine	161
6	Soybean-no till	968	Bare soil	5029	Broadleaf hammock	229
7	Soybean-min till	2468	Bitumen	1330	Hardwood swamp	105
8	Soybean-clean till	614	Bricks	3682	Graminoid marsh	431
9	Woods	1294	Shadows	947	Spartina marsh	520
10					Cattail marsh	404
11					Salt marsh	419
12					Mud flats	503
13					Water	927
Total number of labeled samples		9345		42776		5211

TABLE II
CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS FOR INDIAN PINES DATASET

Class	3DCNN[18]	CDCNN[53]	SSRN[54]	FDSSC[55]	DBMA[38]	DBDA[34]	Baseline	PFSAN	MESAN	ESSAN
1	51.03±13.03	57.02±11.42	78.33±5.45	84.64±6.3	79.68±7.6	83.72±4.94	83.59±3.45	88.03±3.53	89.07±4.69	**89.73±4.27**
2	55.44±5.71	50.11±13.71	90.14±6.63	**95.51±8.55**	85.1±12.65	91.5±9.09	90.76±6.47	94.34±4.94	95.0±4.52	94.63±5.35
3	85.78±5.2	87.56±5.27	95.48±2.48	95.33±2.75	93.15±4.18	94.05±3.18	96.53±2.8	96.92±1.81	96.81±2.37	**97.24±2.43**
4	94.16±4.41	90.66±6.4	99.12±1.37	97.35±1.97	97.92±1.69	97.73±1.42	99.25±0.62	99.41±0.63	99.48±0.67	**99.53±0.56**
5	98.56±1.95	98.06±0.97	99.89±0.22	99.83±0.4	99.32±0.84	99.39±0.6	**100**	**100**	**100**	**100**
6	54.22±12.62	55.66±7.88	85.36±4.89	86.36±6.91	83.2±6.51	84.26±6.61	90.48±2.66	91.2±1.83	91.26±2.58	**91.83±2.78**
7	63.65±10.81	61.92±8.21	78.24±7.71	**92.34±6.39**	78.11±4.17	83.56±5.64	83.58±4.24	84.21±4.52	87.04±4.86	88.04±4.88
8	47.71±6.66	51.54±10.0	88.94±6.57	**97.53±2.28**	88.99±6.22	89.47±5.05	93.3±3.8	96.34±2.49	96.03±3.14	96.2±2.58
9	91.6±3.68	90.5±4.31	98.53±0.94	97.86±3.73	94.81±3.6	96.81±1.76	99.62±0.38	99.57±0.35	99.64±0.32	**99.67±0.31**
OA	68.18±3.0	68.2±1.36	87.19±1.91	92.84±1.57	85.94±1.53	89.05±1.41	90.53±1.63	91.99±0.95	92.97±1.51	**93.41±1.57**
AA	71.35±2.81	71.45±1.82	90.45±1.3	94.08±0.87	88.92±1.6	91.17±1.53	93.01±1.26	94.45±0.61	94.93±1.14	**95.21±1.16**
Kappa	62.68±3.45	62.91±1.47	85.05±2.19	91.6±1.82	83.61±1.78	87.19±1.64	88.95±1.89	90.66±1.09	91.79±1.75	**92.29±1.82**

classification performance in most categories. We also observe that benefiting from our proposed attentions, the recognition ability of the backbone network and discrimination of features extracted are enhanced. Specifically, spatial region distinctions improve Baseline more significantly than weight recalibration of spectral channels, indicating the importance of modeling spatial relationships for the HSI classification task. And the spectral–spatial salient information selection mechanism can bring outstanding performance gains than implementing it in a single domain. Compared to the other two comparison methods with attention mechanisms, DBMA and DBDA, our proposed ESSAN has a clear advantage in terms of classification accuracies. The visualizations in Fig. 7 show that the classification map of Baseline has better homogeneity but more outlier points at the edges of the categories, making the boundaries unclear. This phenomenon is mainly due to the 1×1 convolutional classifier, which does not take into account adjacent pixels when calculating the class probability. The classification maps of PFSAN and MESAN indicate that the proposed attention mechanism can improve boundary misclassifications, while ESSAN improves the network's feature representation capabilities even more on the basis of spectral or spatial attention. The boundary regions of improvements are circled in Fig. 7 with white lines.

2) University of Pavia: It was collected by the ROSIS optical airborne sensor at the University of Pavia, Pavia, Italy. This dataset contains 610×340 pixels and 115 bands covering the wavelength range 430–860 nm with a spatial resolution of 1.3 m, of which 12 noisy bands were removed to retain 103 remaining bands. The dataset covers nine classes of ground objects with a total of 42 776 pixels, including trees, asphalt, bricks, and shadows. It is a typical urban scene. The number of samples available for each category is shown in Table I. We also randomly select 30 and 30 samples per class for training and validation, respectively.

The quantitative and visual results are shown in Table III and Fig. 8, respectively. Our ESSAN still outperforms other

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TABLE III
CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS FOR UNIVERSITY OF PAVIA DATASET

Class	3DCNN[18]	CDCNN[53]	SSRN[54]	FDSSC[55]	DBMA[38]	DBDA[34]	Baseline	PFSAN	MESAN	ESSAN
1	83.53±4.51	83.12±4.56	95.95±3.28	97.79±1.96	95.76±1.98	94.5±2.54	98.02±1.08	97.51±0.86	98.41±1.28	**98.62±0.59**
2	91.45±2.85	91.58±2.71	94.68±4.43	**98.66±1.42**	96.09±1.48	96.0±2.4	97.97±1.98	98.43±1.34	98.31±2.0	98.6±1.59
3	84.98±6.66	79.55±8.2	94.56±3.07	95.16±3.49	92.64±4.51	94.73±2.68	94.59±3.5	95.21±2.4	95.06±3.21	**95.79±4.03**
4	89.18±3.14	94.58±2.48	95.84±2.13	94.12±2.88	93.55±3.97	94.1±3.52	98.16±1.05	**98.21±0.59**	98.2±0.87	98.06±0.98
5	99.57±0.71	99.93±0.16	**100**	99.85±0.34	99.98±0.03	99.89±0.14	99.97±0.05	99.92±0.14	99.91±0.09	99.94±0.13
6	58.22±8.81	74.39±4.79	97.79±1.98	98.24±2.24	97.14±2.93	96.42±3.69	99.23±1.39	98.76±1.38	**99.25±0.76**	98.96±1.69
7	84.3±8.97	87.22±2.61	99.27±0.77	99.97±0.07	98.24±1.79	98.25±1.45	99.64±0.48	99.62±0.48	99.9±0.13	**99.99±0.02**
8	91.91±4.57	76.73±8.67	87.97±10.04	97.26±2.09	91.44±8.45	95.07±2.45	96.95±2.02	96.84±2.33	96.96±2.6	**97.92±1.65**
9	99.17±0.69	99.5±0.35	98.14±5.45	99.31±0.53	99.45±0.41	98.3±1.16	99.69±0.45	99.69±0.39	99.72±0.34	**99.87±0.17**
OA	86.07±1.93	86.9±1.46	95.12±2.07	97.95±0.71	95.67±0.88	95.78±1.17	98.04±0.93	98.13±0.71	98.28±0.92	**98.52±0.81**
AA	86.92±2.12	87.4±1.1	96.02±1.13	97.82±0.65	96.03±0.95	96.36±0.88	98.25±0.55	98.24±0.45	98.41±0.52	**98.64±0.63**
Kappa	81.51±2.54	82.74±1.83	93.6±2.65	97.28±0.93	94.29±1.17	94.44±1.52	97.41±1.22	97.52±0.93	97.73±1.21	**98.05±1.07**

TABLE IV
CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS FOR KENNEDY SPACE CENTER DATASET

Class	3DCNN[18]	CDCNN[53]	SSRN[54]	FDSSC[55]	DBMA[38]	DBDA[34]	Baseline	PFSAN	MESAN	ESSAN
1	91.97±4.34	92.56±4.96	94.15±10.53	90.53±13.44	88.58±9.79	93.44±4.35	**99.68±0.83**	99.57±1.19	99.64±0.88	99.4±1.02
2	81.55±7.73	60.0±10.07	91.76±11.33	80.0±16.54	83.35±11.14	74.21±13.1	90.21±9.84	**92.45±10.09**	90.34±9.83	**92.45±9.35**
3	91.71±3.54	73.17±16.46	86.63±26.01	91.54±10.87	84.51±8.36	79.92±16.2	89.39±5.76	88.54±7.24	90.69±5.18	**92.24±7.38**
4	51.65±16.93	23.39±9.0	51.12±28.77	48.76±30.07	66.94±13.52	52.27±9.51	60.62±12.04	62.36±10.83	67.11±13.2	**69.79±11.78**
5	66.89±18.23	55.36±14.59	**90.79±8.68**	77.42±13.66	71.99±11.78	81.19±14.93	83.51±8.11	82.32±6.87	79.14±7.95	80.66±9.31
6	62.01±8.82	57.58±13.65	61.55±29.53	72.97±30.21	83.84±12.86	90.5±7.65	96.8±4.49	98.08±3.24	**98.54±1.55**	97.72±2.49
7	94.21±5.98	87.68±8.54	87.68±21.57	90.74±13.81	94.42±7.72	93.37±8.22	**100**	**100**	**100**	**100**
8	77.48±13.14	70.86±15.05	86.18±14.45	79.05±21.88	74.61±18.27	81.52±12.39	92.8±10.77	92.99±8.51	**97.41±4.71**	96.03±7.36
9	89.73±5.07	93.69±7.07	95.2±5.23	95.18±7.34	92.88±9.26	92.78±6.64	**96.9±6.08**	96.76±6.38	96.76±6.47	96.82±6.36
10	83.83±7.25	51.29±12.4	97.87±1.93	96.62±4.02	93.73±6.31	93.35±7.15	97.99±5.68	99.64±0.61	**99.87±0.38**	99.04±2.81
11	98.7±2.17	94.08±5.14	99.78±0.4	97.75±2.13	96.72±3.05	96.36±2.9	**99.98±0.07**	99.71±0.88	99.78±0.66	99.88±0.37
12	81.58±7.06	37.0±5.73	91.76±10.39	97.2±3.11	87.02±7.66	81.76±12.08	98.09±2.11	99.01±0.94	98.82±1.35	**99.19±1.23**
13	97.74±1.65	85.97±5.05	**100**	**100.0**	99.17±2.49	**100**	**100**	**100**	**100**	**100**
OA	86.08±1.1	72.78±2.81	91.01±4.36	89.89±3.8	88.61±3.98	88.68±1.7	95.24±1.48	95.58±1.29	96.12±1.02	**96.27±1.43**
AA	82.23±1.68	67.89±3.03	87.27±6.92	85.98±5.32	85.98±3.97	85.44±2.46	92.77±1.75	93.19±1.48	93.7±1.67	**94.09±1.83**
Kappa	84.5±1.23	69.65±3.09	89.98±4.87	88.75±4.22	87.34±4.4	87.39±1.89	94.69±1.65	95.07±1.44	95.68±1.14	**95.84±1.59**

competing methods in terms of numerical results. Similar to the performance of the Indian Pines dataset, the introduction of PFSA or MESA to improve the quality of the representation via spatial or band saliency distinction still provides the advance on the Baseline results. This observation also demonstrates the good adaptability of our proposed attention mechanisms to different types of datasets. The cross-class regions with rich details and complete homogeneous areas are circled in Fig. 8. From the visualization results shown, we can observe that DBMA and DBDA provide smoother visual results. Compared to the soft-attention model, DBDA with self-attention is slightly over-smoothed, with detailed information being covered by large homogeneous areas. Compared to Baseline, ESSAM with spectral–spatial attention mechanism can strike a balance between classification smoothing and spatial detail retention and preserve category boundaries better.

3) Kennedy Space Center: This dataset was acquired by AVIRIS sensor at the Kennedy Space Center, Florida, USA, and contains 512 × 614 pixels and 224 bands in the wavelength range 400–2500 nm with a spatial resolution of 18 m. The 176 bands were retained for the experiment after removing absorption and noise bands. The dataset contains 13 land cover types with a total of 5211 available annotations. Discrimination of land cover for this scene is difficult due to the similarity of spectral signatures for certain vegetation types and the sparsity of ground annotation. The specific category distribution is shown in Table I. We also randomly select ten and ten samples per class for training and validation, respectively.

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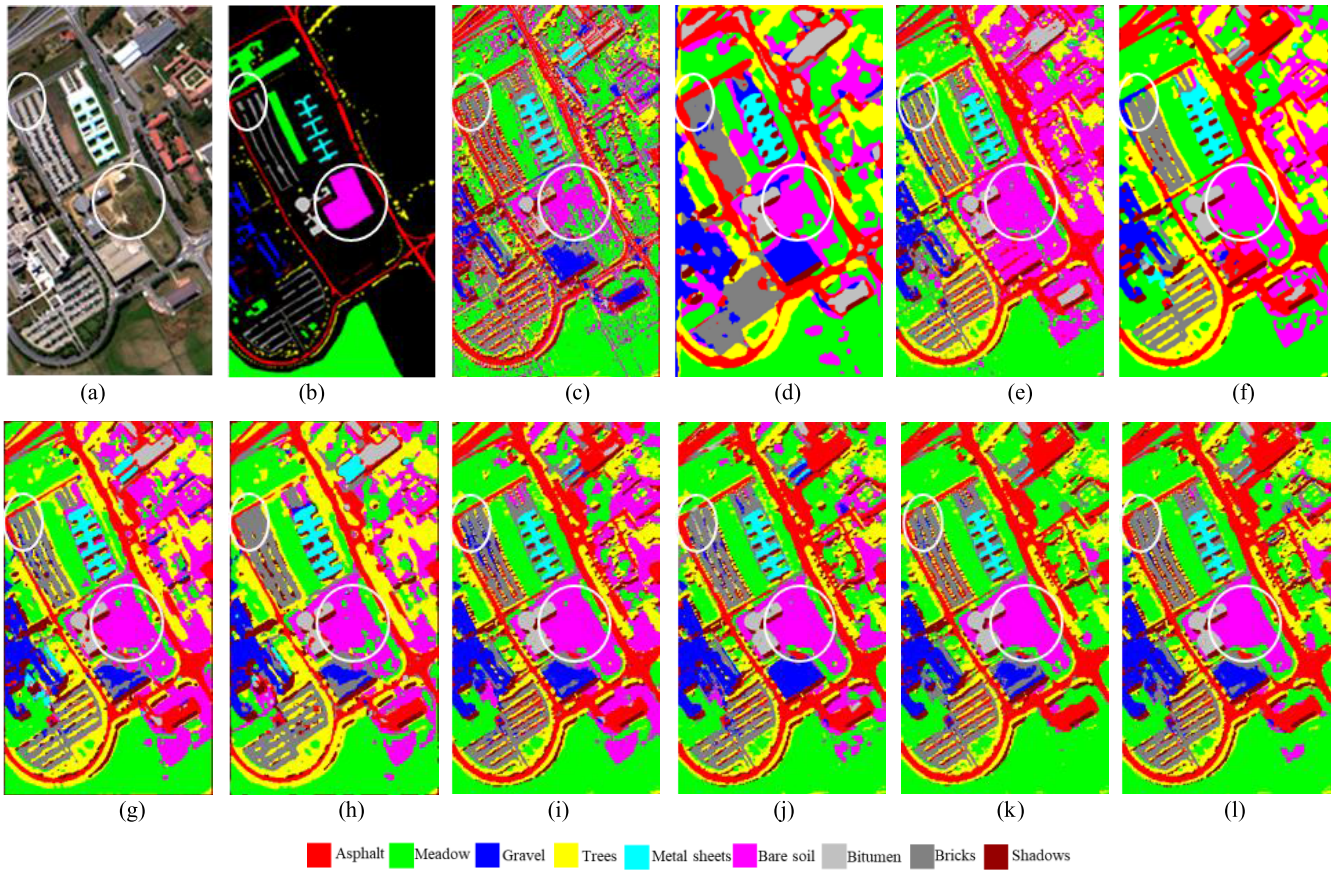

Fig. 8. Classification maps of different methods for the University of Pavia dataset: (a) false-color image; (b) ground-truth map; (c) 3-DCNN; (d) DFFN; (e) SSRN; (f) FDSSC; (g) DBMA; (h) DBDA; (i) Baseline; (j) PFSAN; (k) MESAN; and (l) ESSAN.

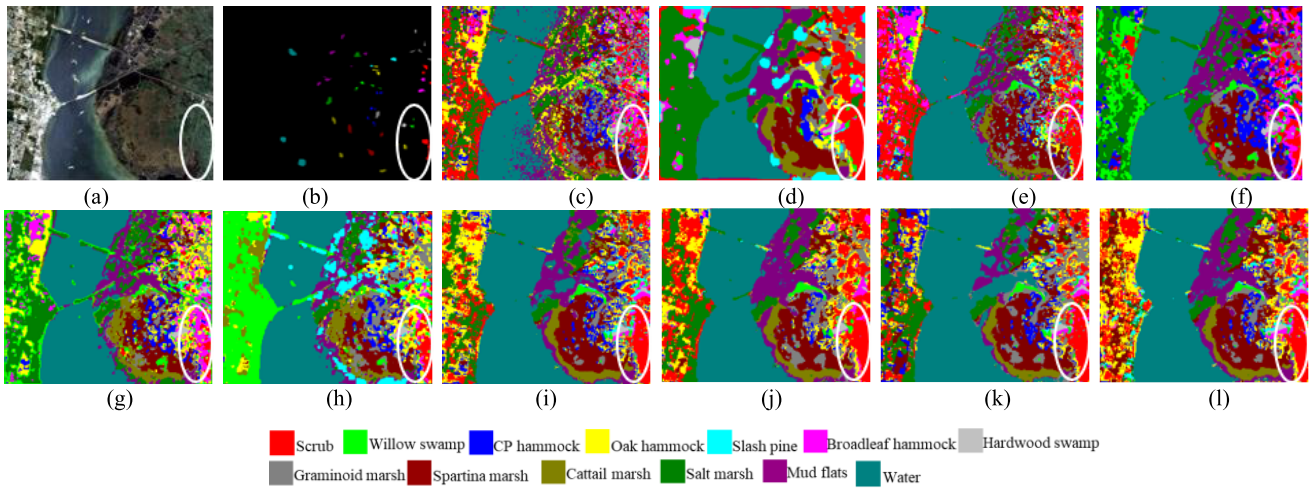

Fig. 9. Classification maps of different methods for the Kennedy Space Center dataset: (a) false-color image; (b) ground-truth map; (c) 3-DCNN; (d) DFFN; (e) SSRN; (f) FDSSC; (g) DBMA; (h) DBDA; (i) Baseline; (j) PFSAN; (k) MESAN; and (l) ESSAN.

The quantitative and visual results are shown in Table IV and Fig. 9, respectively. With less than 3% of the labeled samples, our ESSAN shows a significant advantage over other methods in terms of numerical accuracies. It further demonstrates that our proposed method can be adapted to datasets with different characteristics and provide reliable classification results even in the case of a small number of training samples. Furthermore, the performances of the models adding our attentions are much better than the Baseline, which further validates the effectiveness of our methods. We circle the region in Fig. 9 where ESSAN maintains better category boundaries compared to Baseline.

B. Number of Training Samples

To further analyze and validate the effectiveness and generalization of ESSAN, we conduct controlled experiments with

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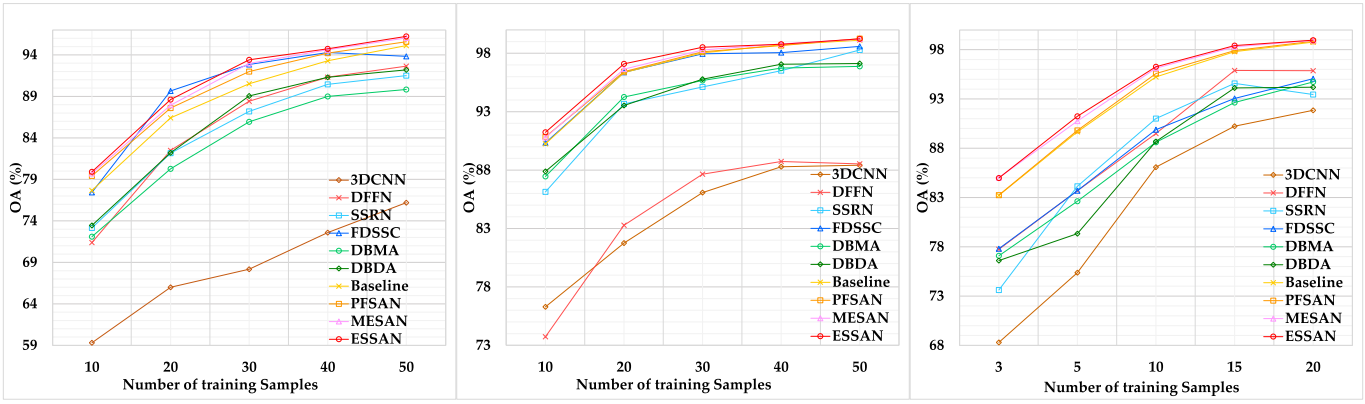

Fig. 10. Comparison of classification performance with different numbers of training samples: (a) Indian Pines; (b) University of Pavia; and (c) Kennedy Space Center.

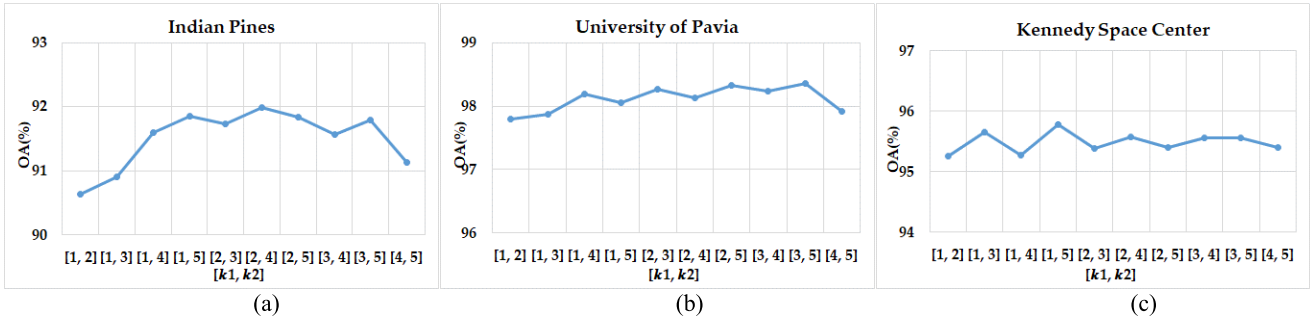

Fig. 11. Comparison of classification performance with different scale control factors: (a) Indian Pines; (b) University of Pavia; and (c) Kennedy Space Center.

different numbers of training samples under three datasets. For the Indian Pines and University of Pavia datasets, 10, 20, 30, 40, and 50 labeled samples of each class are, respectively, selected as the training set, and for the Kennedy Space Center dataset, 3, 5, 10, 15, and 20 samples per class are randomly selected for training. The evaluation results with the OA are shown in Fig. 10. We can observe that classification accuracy of all methods improves with increasing training sample size for any dataset. As the number of training samples decreases, the advantage of our proposed ESSAN relative to competing methods becomes more apparent for OAs on all three datasets. This indicates that ESSAM can still provide high-quality feature representation in the extreme few-sample scenario.

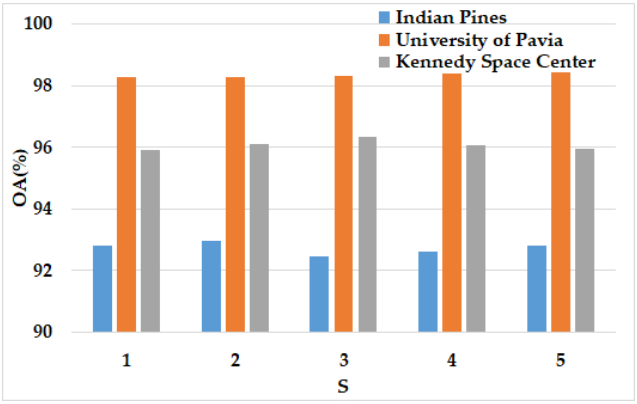

Fig. 12. Comparison of classification performance with different numbers of memory atoms.

C. Hyperparameter Analysis

This section performs validation analysis for the scale control factors of PFSA and the number of memory atoms of LMU of MESA.

Scale Control Factors: For the sake of the balance the computational performance of PFSA and the spectral channel representation capability, the scale factor selection range is set from 1 to 5, from which any two scales k_1 and k_2 are selected to put into one combination $[k_1,k_2]$ for the ablation study. Fig. 11 shows the classification accuracies of PFSAN for different combinations of scales on the three datasets. The experimental results indicate that the Indian Pines dataset is more sensitive to scale variation than the other two datasets, where the difference 6 between the highest and lowest OAs is more than 1%. The above phenomenon is mainly due to the different spatial distribution patterns of the three datasets. The Indian Pines dataset has a more homogeneous and regular spatial distribution of land covers than others, and most categories are arranged in square areas. Combining the classification performance on the three datasets, we chose [2] and [4] as the scale combination for ESSAN.

Number of Memory Atoms: Since the aim of LMU is to capture the category potential subspace of the image, the number of memory atoms for different datasets is set to be an integer multiple of the number of classes in that dataset. Fig. 12 reports the classification accuracies of MESAN at

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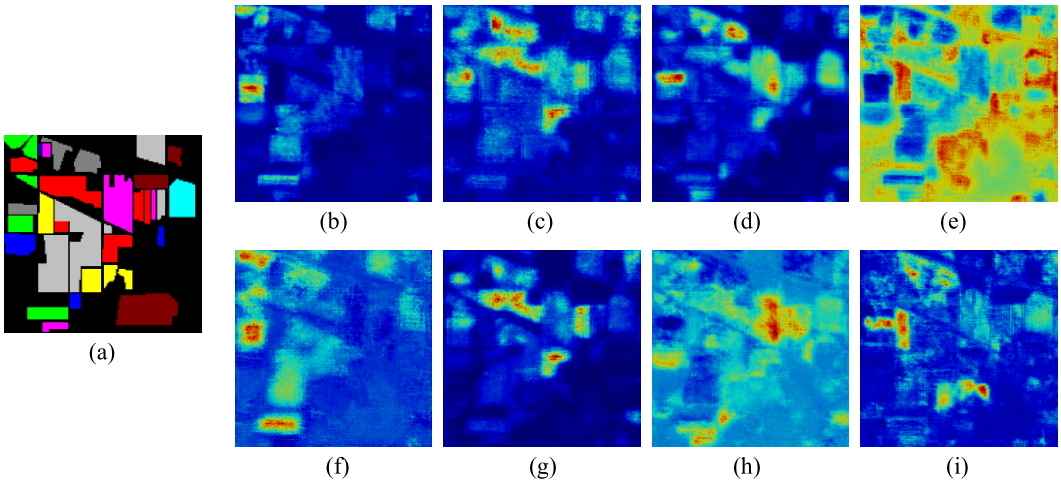

Fig. 13. Visualization of relationship matrices for Indian Pines: (a) ground-truth map; (b)-(e) relationship graphs between pixels and memory atoms of MESAN; and (f)–(i) relationship graphs between pixels and memory atoms of ESSAN.

different numbers of atoms, where S means the multiple ranging from 1 to 5. It can be observed that the number of atoms has the least effect on the classification accuracy of the University of Pavia dataset, whose OA increasing as the number of atoms increases. The Indian Pines dataset is the most sensitive to the number of atoms, and the best performance is achieved when the number of atoms was twice the number of categories. The optimal settings are not the same for different datasets. However, considering the balance between the classification performance and the increase in the number of parameters, we set the number of atoms in the LMU to be twice the number of categories in each dataset.

D. Performance Comparison With Classical Attention Mechanisms

To further validate the superiority of the proposed attention mechanisms, we design corresponding ablation experiments that compare our ESSAM with multiple classical attention mechanisms, i.e., SE block, ECA block, and CBAM. Specifically, the ESSAM of ESSAN is replaced with each of the classical modules mentioned above, which are combined with the backbone network to construct the corresponding networks. In this case, the specific settings of these classical attention models follow the relevant references. The reduction ratios of SE block and CBAM are both 16. The kernel size of ECA block is 5. The kernel size of CBAM is 7×7. The classification accuracies are shown in Table V.

It can be seen that the proposed spectral–spatial attention mechanism ESSAM provides maximum classification performance gain on all three datasets with higher effectiveness than the classical attention modules. Specifically, the overall accuracies of ESSAN for Indian Pines and Kennedy Space Center are obvious advance in Baseline with SE, ECA, and CBAM. For the University of Pavia dataset, the accuracy of ESSAN is slightly improved compared to "Baseline + CBAM," the best performing of the classical attentions. In general, feature refinement in the spectral–spatial domain is more effective than in a single dimension.

TABLE V
CLASSIFICATION ACCURACIES (%) WITH DIFFERENT ATTENTION MODELS

	Indian Pines	University of Pavia	Kennedy Space Center
Baseline	90.53	98.04	95.24
Baseline+SE ([27])	91.01	97.49	95.42
Baseline+ECA ([40])	<u>91.67</u>	98.19	95.42
Baseline+CBAM ([39])	91.61	<u>98.47</u>	<u>95.55</u>
ESSAN	**93.41**	**98.52**	**96.27**

E. Effectiveness Analysis of Memory Enhancement

Relational inference, an important aspect of UPA, has a direct impact on the classification performance of the model in terms of the accuracy of its information transfer. To assess the learning ability of LMU in MESA for potential representations of images, the relational matrices are visualized based on (22), which belong to partial atoms for Indian Pines and University of Pavia and are displayed in Figs. 13 and 14.

Fig. 13(a) is the ground-truth map for the Indian Pines dataset, Fig. 13(b)–(e) are the spatial correlations captured by MESAN, and Fig. 13(f)–(i) are the spatial correlations in ESSAN after spectral channel refinement. The four columns of relationship maps in Fig. 13 show the long-range dependencies for the four classes of Corn-min ill, Corn-no till, Soybean-no till, and Grass-Trees, from left to right. It is easy to see that the spatial relations of MESA are category-based, i.e., the LMU is served as a medium to establish spatial dependencies of pixels in the same category, which is not limited to the local neighborhood. In other words, LMU adaptively captures the semantic encoding of each category in a learnable manner. A longitudinal comparison indicates that the spectralwise recalibration of ESSAN leads to a more accurate localization of spatial relations, indirectly demonstrating the effectiveness of PFSA.

Fig. 14(a) is the ground-truth map for University of Pavia, Fig. 14(b)–(e) belong to MESAN, and Fig. 14(f)–(i) belong

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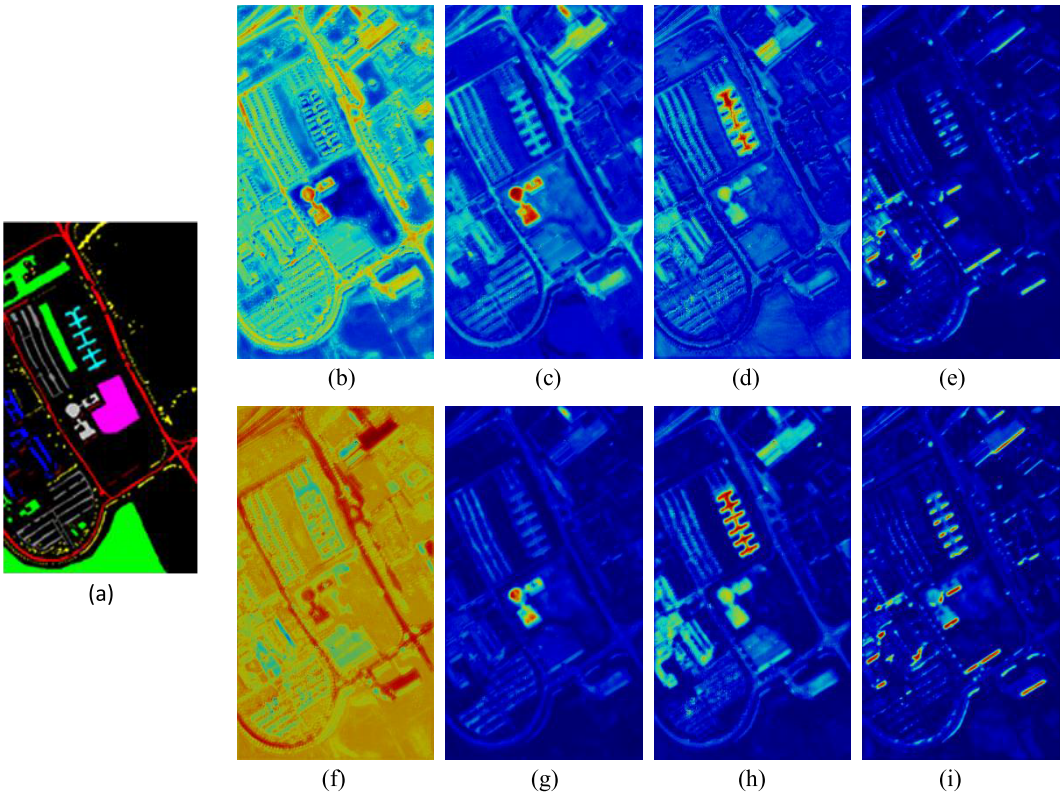

Fig. 14. Visualization of relationship matrices for the University of Pavia: (a) ground-truth map; (b)–(e) relationship graphs between pixels and memory atoms of MESAN; and (f)–(i) relationship graphs between pixels and memory atoms of ESSAN.

to ESSAN. The global correlations for the four categories of asphalt, bitumen, metal sheets, and shadows are established in Fig. 14(b) and (f), Fig. 14(c) and (g), Fig. 14(d) and (h), and Fig. 14(e) and (i), respectively. Unlike Indian Pines, the distribution of land covers in the University of Pavia dataset is not regular, with most categories being distributed in narrow bands. It can be seen that MESA still accurately establishes spatial associations of similar pixels to perform information transfer and feature enhancement through the corresponding atoms of LMU.

V. Conclusion

In this article, we proposed a UAP with the corresponding network framework and instantiated the ESSAM suitable for HSI classification task to achieve the feature refinement in the global scope. The spectral attention exploits the spatial region distribution of HSIs to extract multiscale structured representations and perform parameter-free cross-channel interaction. The spatial attention mechanism employs a set of memory atoms to capture key semantic information of HSIs to achieve the efficient spatial relationship modeling. ESSAM takes full advantage of the structural priors of HSIs and provides an efficient scheme with fewer parameters to explicitly establish the correlations between all channels and all pixels and perform dynamic feature recalibration of spectral and spatial dimensions. The experimental results from three public datasets demonstrate that the proposed ESSAN is capable of delivering SOTA performances for HSI classification. Additionally, ESSAM exhibits greater adaptability to this task than other classical attention modules. In addition, the effectiveness of ESSAM also demonstrates the validity of UPA, which also provides a new strategy for the design of attention modules.

The purpose of deep neural networks is finding the discriminative low-dimensional representation by analyzing the correlation of high-dimensional data, which is highly relevant to SR methods. In the future, we will investigate the possible combination of DL and SR and develop more DL methods to explore low-dimensional structure of HSIs and further improve the efficiency, performance, and interpretability of the DL models.

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